Concentric Zero-Spheres of a class of sections of the commutative ring $\mathcal{R}$ of slice preserving functions over the compactified Octonions

with the Riemann Hypothesis as a corollary

Jesse Michael Paul
PhD Candidate, Department of Mathematics & Statistics
University of North Carolina at Greensboro

June 2026

"...a 'viewpoint' by itself remains fragmentary. It reveals to us one of the aspects of a scenery or panorama, among a multiplicity of others which are equally valuable, equally 'real'. It is when complementary viewpoints of a common reality are conjugated, that is, when our 'eyes' are multiplied, that the gaze is able to penetrate further ahead in the reckoning of things. ...it also happens, sometimes, that a sheaf of viewpoints converging to a unique and vast scenery gives rise to a novel thing; a thing which transcends each of the partial perspectives, in the same way that a living being transcends each of its limbs and organs. This new thing could be called a vision."

— A. Grothendieck, *Récoltes et Semailles*

Abstract

Let $\mathcal{R}$ be the commutative ring of slice-preserving slice-regular functions on the compactified octonions $\mathbb{O}^* = S^8$, under the regular $(*)$ product. Part 1 gathers the classical facts about the Riemann zeta function. Part 2 develops the slice-preserving theory, builds $\mathcal{R}$, places the octonionic zeta $\zeta_{\mathbb{O}^*}$ inside it, and proves the zero and equivalence theorems: the classically nontrivial zeros of ζ are residue-$\mathbb{C}$ zero 6-spheres, and the Riemann Hypothesis holds exactly when those spheres are concentric, sharing a single real centre.

Part 3 proves the main result, the *Concentricity Theorem*, for a section $A \in \mathcal{R}$ satisfying four conditions: (C1) meromorphic continuation with a single simple pole carried to $\infty = N$; (C2) an infinite Euler product; (C3) an infinite slice-regular Weierstrass factorization whose spherical factors are the residue-$\mathbb{C}$ zeros; and (C4) infinitely many such zeros. The theorem says the residue-$\mathbb{C}$ zero 6-spheres of any such A are concentric.

The proof builds a concentricity detector from the section itself, out of groupoids, orbit–stabilizer, and categorical homotopy theory. Conditions (C1)–(C3) determine one distinguished exponential Möbius action on the projective base $\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R})$: a logarithmic coordinate z is exponentiated in the diagonal matrix $\mathrm{diag}(e^z, 1)$ and conjugated into the Cayley disk chart, with the Euler product presenting the action at 0 and the Weierstrass product at N. The slice-preserving normalization is G_2-equivariant on $\mathbb{O}^*$, and orbit–stabilizer carries this whole point-valued action across the base. Its sphere-world projection records the directions and Möbius legs, while the genuine A-section functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$ has the resulting normalized A-value states and their transports as its fibres. Its Grothendieck construction is $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$.

The C-residue system is the inverse-image action groupoid $\mathcal{R}_A$ selected inside $\mathcal{A}_A$, with fully faithful inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$, and hence a fully faithful inclusion of totals $\int \iota_A : \int_{\mathcal{B}} \mathcal{R}_A \hookrightarrow$

$\mathcal{T}_A$. The relative orbit–stabilizer comparison makes this inverse-image total transitive, therefore connected. Its component colimit is consequently a singleton. Instantiated at any two certified residue representatives, that singleton equality is already equality of their A-specific real-level reads; hence all of them have one real value $c \in \mathbb{R}$. Thus every populated residue-$\mathbb{C}$ zero 6-sphere has centre c. The theorem asserts the shared centre; naming the number is the task of the corollaries.

Downstream corollaries translate residue-$\mathbb{C}$ zeros to classically nontrivial zeros and, for $\zeta_{\mathbb{O}^*}$, use its functional equation to identify $c = \frac{1}{2}$: the Riemann Hypothesis.

Contents

Introduction

The Riemann Hypothesis is usually met head-on, as a question about where the zeros of a single complex function fall along a line. This paper approaches it from the side. It asks instead whether a family of geometric objects, sitting a few dimensions up in the octonions, share a common centre, and shows that they must. The Riemann Hypothesis then follows, not as the thing we set out to prove, but as the shadow this larger fact casts back down onto the complex plane.

A nontrivial zero of the zeta function, seen from the complex line alone, is a lone point in a strip. Seen from the compactified octonions, that same zero opens into a six-dimensional sphere, and the critical-line question becomes a question of arrangement: are these spheres concentric, or scattered into different hyperplanes? It takes the whole continuum of imaginary directions in the octonions, read together, to see whether the alignment holds.

The paper is built to make that convergence visible and, at every step, checkable. Part 1 recalls the classical facts about the Riemann zeta function we will lean on. Part 2 develops the slice-preserving theory of functions on the octonions, builds the commutative ring $\mathcal{R}$ in which the octonionic zeta lives, and proves the reframing at the centre of everything: the classically nontrivial zeros are exactly the residue-$\mathbb{C}$ zero six-spheres, and the Riemann Hypothesis holds precisely when those spheres are concentric. Part 3 proves the main result, the Concentricity Theorem, with a small piece of machinery I think of as a concentricity detector. From the section itself we build one distinguished action on a projective base, carry it across a continuum of Riemann spheres by orbit and stabilizer, and read off, through categorical homotopy theory, whether the zero-spheres land in a single connected piece. They do; and for the octonionic zeta, that is the Riemann Hypothesis.

No one can picture an eight-dimensional graph in their head, but tools from categorical homotopy theory, like the total-object Grothendieck construction and standard colimit arguments, let us probe this structure and reason about it. The argument is also formalized in Lean, so that each step can be checked; the reader can not only verify the steps but, I hope, come to understand for themselves why the Concentricity Theorem is true and why the Riemann Hypothesis follows.

Part I
Classical Background

1 The Riemann zeta function

The Riemann zeta function is classical, and we take its basic theory as given. We recall only what the construction uses: the meromorphic continuation and functional equation, the Euler product over the primes, the Hadamard product over the zeros, and Hardy's theorem. We then read ζ in compactified form, as a map to the Riemann sphere, which is the form in which it will extend to the octonions.

Theorem 1.1 (Meromorphic continuation and functional equation; Riemann [13])**.** *The Dirichlet series $\zeta(s) = \sum_{n\geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$ with a single simple pole, at $s = 1$; equivalently, a holomorphic map to the Riemann sphere*

$$\zeta : \mathbb{C}^* \longrightarrow \mathbb{C}^*, \qquad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \qquad \zeta(1) = \infty.$$

The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies

$$\xi(s) = \xi(1-s), \qquad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ is a nontrivial zero of ζ then so are $\bar{\rho}$, $1-\rho$, and $1-\bar{\rho}$. The trivial zeros of ζ are at $s = -2, -4, \ldots$; the nontrivial zeros lie in the strip $0 < \operatorname{Re} s < 1$. The Riemann Hypothesis asserts that every nontrivial zero has $\operatorname{Re} s = \frac{1}{2}$.

Theorem 1.2 (Infinite Euler product; [15, Ch. 1])**.** *For $\operatorname{Re} s > 1$,*

$$\zeta(s)=\prod_{p\ \text{prime}}\left(1-p^{-s}\right)^{-1}=\exp\Bigl(\sum_p\sum_{k\ge 1}\tfrac{1}{k}\,p^{-ks}\Bigr),$$

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\operatorname{Re}s>1$*.*

Theorem 1.3 (Infinite Hadamard product; [15, Ch. 2])**.** *ξ is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros $\{\rho\}$ of ζ, and admits the Hadamard factorization*

$$\xi(s)=\xi(0)\prod_\rho\Bigl(1-\tfrac{s}{\rho}\Bigr)e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Corollary 1.4 (Infinitude of the nontrivial zeros)**.** *ζ has infinitely many nontrivial zeros.*

Proof. Immediate from Theorem 1.3: its zeros are infinitely many and are exactly the nontrivial zeros of ζ. □

Theorem 1.5 (Hardy [11])**.** *Infinitely many nontrivial zeros of ζ lie on the line* $\operatorname{Re}s=\frac{1}{2}$*.*

The compactified Riemann zeta on $\mathbb{C}^*$

For the slice construction, ζ is taken in compactified form, with values on the Riemann sphere $\mathbb{C}^*=\mathbb{C}\cup\{\infty\}=S^2$ (the one-point compactification of $\mathbb{C}$ [12]). The classical facts above are retained as stated.

Definition 1.6 (Compactified classical zeta)**.** $\zeta_{\mathbb{C}^*}:\mathbb{C}^*\to\mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s)=\zeta(s)$ for $s\in\mathbb{C}\setminus\{1\}$; $\zeta_{\mathbb{C}^*}(1)=\infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty)=1$, since $\zeta(s)\to 1$ as $\operatorname{Re}(s)\to+\infty$ (the p-test: only the $n=1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C}^*\setminus\{1\}$, its sole pole being $s=1$. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s})=\overline{\zeta_{\mathbb{C}^*}(s)}$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C}\setminus\{1\}$ where $\zeta_{\mathbb{C}^*}=\zeta$.

Lemma 1.7 (Compactification preserves the zero set)**.** *$Z(\zeta_{\mathbb{C}^*})=Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.*

Proof. On $\mathbb{C}\setminus\{1\}$ the two functions agree, so they share all zeros there. The two adjoined values lie outside the zero set: $\zeta_{\mathbb{C}^*}(1)=\infty$ is a pole and $\zeta_{\mathbb{C}^*}(\infty)=1$. Hence compactification preserves the zero set exactly. □

Part II
Slice-Preserving Theory, the Ring $\mathcal{R}$, and the Equivalence

2 Octonions and slices

The octonions are an 8-dimensional real division algebra, the largest of the normed division algebras. This section fixes what we need from them: the algebra itself; Artin's theorem, that any two octonions generate an associative subalgebra, so that powers and slicewise complex analysis make sense; and the complex slices $\mathbb{C}_v$, one for each imaginary direction $v \in S^6$, together with their compactifications, the slice Riemann spheres. All the slices share the real axis and a single point at infinity, the north pole N; that shared geometry is what the later construction turns around.

The octonions $\mathbb{O}$ have basis $\{1, e_1, \ldots, e_7\}$ with each $e_i^2 = -1$.

Definition 2.1. The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

An octonion $s \in \mathbb{O}$ decomposes as $s = \sigma + w$ with $\sigma = \mathrm{Re}(s) \in \mathbb{R}$ and $w = \mathrm{Im}(s) \in \mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$; the unit imaginary octonions form $S^6 \subset \mathrm{Im}(\mathbb{O})$.

Theorem 2.2 (Artin [14])**.** *In an alternative algebra, the subalgebra generated by any two elements is associative.*

Corollary 2.3. *Powers s^n are well-defined for any $s \in \mathbb{O}$.*

Definition 2.4 (Complex slices and slice Riemann spheres)**.** For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$-algebra isomorphism $\varphi_v : \mathbb{C} \to \mathbb{C}_v$, $i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

3 Slice-preserving functions

The class we use is the *slice-preserving* slice-regular functions. We first recall slice regularity [8, 6, 5, 9], holomorphicity on each complex slice separately, as the ambient notion, then isolate the slice-preserving subclass, on which classical complex methods apply slicewise (by Artin's theorem each slice $\mathbb{C}_v$ is an associative subalgebra where classical complex analysis applies unchanged).

Definition 3.1 (Slice regularity; [5, Def. 5.1.1])**.** Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \to \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

Theorem 3.2 (Representation Formula; [5, Thm. 5.1.7])**.** *Let f be slice regular on an axially symmetric slice domain Ω. Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2} v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.*

Theorem 3.3 (Identity Theorem; [5, Cor. 5.1.9])**.** *Let f be slice regular on an axially symmetric slice domain Ω. If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω.*

Theorem 3.4 (Extension Theorem; [5, Thm. 5.1.5])**.** *Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \to \mathbb{C}$ holomorphic. There is a unique slice regular $\mathrm{ext}(g) : \Omega_U \to \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U,\ v \in S^6\}$ with $\mathrm{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0.*

Definition 3.5 (Slice-preserving functions; [1, Def. 2.7, Rem. 2.8]; [17, 7, 16])**.** A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

(i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);

(ii) its representation-formula components are $\mathbb{R}$-valued: writing $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ (Theorem 3.2), one has $\alpha, \beta : \mathbb{R}^2 \to \mathbb{R}$, equivalently the inducing stem function $F = F_1 + \iota F_2$ has $\mathbb{R}$-valued components F_1, F_2;

(iii) f preserves two distinct slices; equivalently $f(\Omega \cap \mathbb{R}) \subseteq \mathbb{R}$ when Ω is a slice domain.

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti– Stoppato [7, Rem. 1.8]).

Remark 3.6 (Compactified extension)*.* Definition 2.4 through Definition 3.5, and Theorems 3.2–3.4, are stated in the literature on uncompactified domains $\Omega \subseteq \mathbb{O}$ (and $U \subseteq \mathbb{C}$). We use them on the compactified $\mathbb{O}^* = S^8$ and on the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (GPV's slice Riemann manifolds [17]). The passage $\mathbb{O} \rightsquigarrow \mathbb{O}^*$, $\mathbb{C}_v \rightsquigarrow \mathbb{C}_v^*$ adjoins only the shared point $\infty = N$; it is recorded as a marked extension node (in the Lean development: cite the uncompactified statement and extend through the one-point compactification `OnePoint`).

4 The octonionic zeta function $\zeta_{\mathbb{O}^*}$

Slice-preserving theory already gives the tools to build the octonionic zeta function. On each slice $\mathbb{C}_v$ we read the compactified classical ζ through the isomorphism $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^*$, and the representation formula glues these slice copies into a single function on all of $\mathbb{O}^*$. The one thing to check is that the two identifications of a slice with $\mathbb{C}$ agree, so that $\zeta_{\mathbb{O}^*}$ is well-defined; they do, because conjugating the input and reversing the identification cancel. The pole of ζ at $s = 1$ is carried to the north pole N.

Definition 4.1 (Octonionic zeta)**.** Here N denotes the *north pole* of S^8: the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty = N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^* = S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \to \mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^* = S_v^2$ (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty) = N$; GPV [17], Prop. 4.1/Thm. 4.2):

(i) for $s \in \mathbb{O} \setminus \mathbb{R}$, with $w = \operatorname{Im}(s)$, $v = w/|w| \in S^6$, $\gamma = |w| > 0$, $\sigma = \operatorname{Re}(s)$, so $s = \sigma + \gamma v$: $\zeta_{\mathbb{O}^*}(s) := \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$;

(ii) for $s \in \mathbb{R} \setminus \{1\}$: $\zeta_{\mathbb{O}^*}(s) := \zeta_{\mathbb{C}^*}(s) = \zeta(s) \in \mathbb{R}$;

(iii) for $s = 1$ (the simple pole, Definition 1.6): $\zeta_{\mathbb{O}^*}(1) := \infty = N$;

(iv) for $s = \infty = N$: $\quad \zeta_{\mathbb{O}^*}(\infty) := \zeta_{\mathbb{C}^*}(\infty) = 1$.

Proposition 4.2 (Well-definedness)**.** *$\zeta_{\mathbb{O}^*} : \mathbb{O}^* \to \mathbb{O}^*$ is well-defined.*

Proof. For $s \in \mathbb{O} \setminus \mathbb{R}$ the decomposition $s = \sigma + \gamma v$ with $\gamma > 0$, $v \in S^6$ is forced ($\gamma = |\operatorname{Im} s|$, $v = \operatorname{Im} s/|\operatorname{Im} s|$), so the only ambiguity is the identification of the slice $\mathbb{C}_v = \mathbb{C}_{-v}$ with $\mathbb{C}^*$. That slice carries two such identifications, φ_v ($i \mapsto v$) and φ_{-v} ($i \mapsto -v$), under which s reads as $\sigma + i\gamma$ resp. $\sigma - i\gamma$; we must check $\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)) = \varphi_{-v}(\zeta_{\mathbb{C}^*}(\sigma - i\gamma))$. By conjugate symmetry $\zeta_{\mathbb{C}^*}(\sigma - i\gamma) = \overline{\zeta_{\mathbb{C}^*}(\sigma + i\gamma)}$ ([15, Ch. 2]; Definition 1.6), writing $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = a + bi$ gives $\varphi_{-v}(a - bi) = a + (-b)(-v) = a + bv = \varphi_v(a + bi)$: the two sign reversals, one from conjugating the input, one from the reversed identification, cancel. The remaining cases carry no ambiguity: for real $s \neq 1$, $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$ for every v; at $s = 1$ the value is the single point $\infty = N$; and at $s = \infty = N$ the value is $\zeta_{\mathbb{C}^*}(\infty) = 1 \in \mathbb{R}$, again the same for every v since each φ_v fixes $\mathbb{R}$ pointwise. ☐

5 $\mathcal{R}$ is a commutative ring

The slice-preserving sections form a commutative ring $\mathcal{R}$, with $\zeta_{\mathbb{O}^*}$ among its elements. Its multiplication is the regular $(*)$ product, which descends from the Cayley–Dickson construction of $\mathbb{O}$ and from slice regularity; on slice-preserving functions it restricts on each slice to the ordinary pointwise product (Wang), and is therefore commutative. The ring axioms, and the integral-domain property, follow slicewise.

We work on $\mathbb{O}^*$; write $\Omega_v := \mathbb{O}^* \cap \mathbb{C}_v = \mathbb{C}_v^* = S_v^2$ for the slice Riemann sphere (Definition 2.4).

Definition 5.1 (The ring of slice-preserving functions under the regular product)**.** $\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-regular functions $f : \mathbb{O}^* \to \mathbb{O}^*$ that are slice preserving (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, equipped with pointwise addition and the regular $(*)$ product.

Commutativity of $\mathcal{R}$ rests on a single slice-preserving fact from the literature: on each slice the regular product is ordinary multiplication.

Theorem 5.2 (Regular product on slice-preserving functions; [16, Rem. 2.11])**.** *For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).*

Proposition 5.3. *$(\mathcal{R}, +, *)$ is a commutative ring with identity $1 \in \mathcal{R}$.*

Proof. Closure under $+$ is pointwise. For $*$: fixing a slice $\mathbb{C}_K$, the product reduces to pointwise multiplication on $\mathbb{C}_K$ by Theorem 5.2, which is commutative because $\mathbb{C}_K$ is a commutative associative subalgebra; associativity and distributivity are inherited slicewise, and a slice-regular function is determined by its restriction to any one slice (Theorem 3.3). The constant 1 is the identity. ☐

Proposition 5.4 ($\mathcal{R}$ is an integral domain)**.** *If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.*

Proof. $(f * g)|_{\Omega_K}(w) = f(w) \cdot g(w)$ in $\mathbb{C}_K \cong \mathbb{C}$ (Theorem 5.2). The restrictions $f|_{\Omega_K}$, $g|_{\Omega_K}$ are holomorphic on the connected slice; by the classical identity principle the pointwise product vanishes identically iff one factor does. By Theorem 3.3 the corresponding global function vanishes. ☐

6 The slice-preserving analytic toolkit on the compactified octonions

We collect the slice-preserving analytic facts used in Part 3 as cited statements: the exponential, its logarithm manifold, its degenerate fibre, the slice/stem square and I-independent lift, and the winding lift. Each is established in the literature on the uncompactified $\mathbb{O}$ and used here on $\mathbb{O}^* = S^8$ and the slice Riemann spheres $\mathbb{C}_v^* = S_v^2$ (the passage is recorded in Remark 3.6; the compactified slice-manifold structure itself is GPV's [17]).

Theorem 6.1 (The slice exponential; [21, §3]; [17, Prop. 5.1]; [1, Rem. 2.23])**.** *The exponential* $\exp : \mathbb{O} \to \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k\geq 0} q^k/k!$ *(powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by*

$$\exp(x + Iy) = e^x(\cos y + I \sin y), \qquad x, y \in \mathbb{R},\ I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ *depends only on the real part, and* $\exp$ *is nowhere zero.*

Theorem 6.2 (The logarithm manifold $E^+_{\mathbb{O}}$; [17, Prop. 5.1, Rem. 5.2, Def. 5.3, Prop. 5.4, Def. 5.5]; [21, §3])**.** *Let* $T : \mathbb{O} \to \mathbb{O} \times \operatorname{Im} \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y,\ Iy)$, *and* $\mathbb{O}^+ = \{q : \operatorname{Re} q > 0\}$*. The* logarithm manifold *is* $E^+_{\mathbb{O}} := T(\mathbb{O}^+)$*, the semi-helicoidal hypercomplex 8-manifold; equivalently it is the image of the exponential graph, parametrised by the* $E^+_{\mathbb{O}}$-exponential

$$E : \mathbb{O} \longrightarrow E^+_{\mathbb{O}} \subset \mathbb{O} \times \operatorname{Im} \mathbb{O}, \qquad E(x + Iy) = (\exp(x + Iy),\ Iy) = (e^x \cos y + I e^x \sin y,\ Iy),$$

an immersion and a diffeomorphism, with inverse the $E^+_{\mathbb{O}}$-logarithm $L : E^+_{\mathbb{O}} \to \mathbb{O}$, $L(q, p) = \log|q| + p$ *(here* $p = \operatorname{Arg}(q)$*). Writing* π *for the projection to the first factor,* $\pi \circ E = \exp$ *(Theorem 6.1). The manifold* $E^+_{\mathbb{O}}$ *is an adapted blow-up of* $\mathbb{O}$ *along the real axis. Its projection* $\pi : E^+_{\mathbb{O}} \to \mathbb{O} \setminus \{0\}$ *has the degenerate real fibres responsible for the failure of covering and openness. The blow-up unwraps the multivalued logarithm into the single-valued* L*, and a branch of the hypercomplex logarithm is* $\log_{\mathbb{O}} = L \circ (\pi|_\Omega)^{-1}$ *on any path-connected* $\Omega \subset E^+_{\mathbb{O}}$ *on which* $\pi|_\Omega$ *is injective.*

Lemma 6.3 (The degenerate set of the exponential; its fibre over a real value)**.** Sourced. *With* π *the first-factor projection,* $\pi \circ E = \exp$ ([17, Rem. 5.2(a)], the commuting triangle)*. Its spherical* degenerate set *makes the projection an adapted blow-up with failures of covering and openness* ([17, Rem. 5.2(b)])*. The direction* $I(q)$ *becomes singular at every point of* $\mathbb{R}$ ([21, Rem. 2.1])*.*

Proved from the slice form. *For* $r > 0$*,*

$$\exp^{-1}(-r) = \big\{ \log r + I(2k+1)\pi :\ I \in S^6,\ k \in \mathbb{Z} \big\} :$$

by Theorem 6.1, $\exp(x+Iy) = e^x \cos y + I e^x \sin y = -r$ *forces* $e^x \sin y = 0$*, hence* $y \in \pi\mathbb{Z}$*; negativity of the value forces* $y = (2k+1)\pi$ *and* $x = \log r$*; the direction* I *is unconstrained. The fibre is thus indexed by the single real* level $\log r = \log|-r|$ *and, within it, by the odd winding indices* $2k+1$ *carried as band data (the winding lift, Theorem 6.5; [21, Cor. 5.21]). The fibre formula itself appears, stated without proof as Preface motivation, in [17] (p. 972); the slice-form derivation above is the load-bearing statement.*

Definition 6.4 (Slice restriction and the I-independent lift; [16, Rem. 2.11]; [4, §3.3, §3.7]; [7, Def. 2.1, Prop. 2.2])**.** A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \qquad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \ \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ of Definition 2.4. The *I-independent lift* is the fact that this holomorphic stem is the *same* for every v: there is one holomorphic $A_\circ : \mathbb{C}^* \to \mathbb{C}^*$ with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_\circ \circ \varphi_v^{-1} \qquad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \to \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$-valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I\, F_2(x + iy), \qquad F_1, F_2 \ \mathbb{R}\text{-valued}.$$

It is this single I-independent slice stem that makes a slice-preserving section G_2-equivariant, hence blind to the G_2-orbit direction (Definition 9.2).

Theorem 6.5 (The winding lift; [21, Def. 3.4, Def. 4.1, Prop. 4.2, Def. 5.11])**.** *Let $\gamma : [a, b] \to \mathbb{O}\setminus\{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation). A* continuation of the hypercomplex logarithm along γ *is a path $\tilde{\gamma} : [a, b] \to \mathbb{O}$ with $\exp \circ \tilde{\gamma} = \gamma$, and a* lift of γ *to the logarithm manifold $E^+_{\mathbb{O}}$ (Theorem 6.2) is a path $\Gamma : [a, b] \to E^+_{\mathbb{O}}$ with $\mathrm{pr}_1 \circ \Gamma = \gamma$:*

$$\begin{tikzcd} & \mathbb{O} \\ {[a,b]} \arrow[ur, "\tilde{\gamma}"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \arrow[u, "\exp"'] \end{tikzcd} \qquad \begin{tikzcd} & E^+_{\mathbb{O}} \arrow[d, "\mathrm{pr}_1"] \\ {[a,b]} \arrow[ur, "\Gamma"] \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

The two data are equivalent*: a lift Γ exists if and only if a continuation $\tilde{\gamma}$ exists, and they correspond by*

$$\tilde{\gamma} = L \circ \Gamma, \qquad \Gamma = (\exp \circ \tilde{\gamma},\ \operatorname{Im} \tilde{\gamma}),$$

where L is the $E^+_{\mathbb{O}}$-logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a loop *$\gamma : S^1 \to \mathbb{O} \setminus \{0\}$ the lift is taken on the universal cover $\exp : i\mathbb{R} \to S^1$, as a continuous $\Gamma : i\mathbb{R} \to E^+_{\mathbb{O}}$ with $\mathrm{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):*

$$\begin{tikzcd} i\mathbb{R} \arrow[r, "\Gamma"] \arrow[d, "\exp"'] & E^+_{\mathbb{O}} \arrow[d, "\mathrm{pr}_1"] \\ S^1 \arrow[r, "\gamma"'] & \mathbb{O} \setminus \{0\} \end{tikzcd}$$

Proposition 6.6 (Tameness, existence, and the winding number; [21, Def. 4.20, Cor. 5.21, Cor. 5.22])**.** *The lift of Theorem 6.5 is governed by the companion structure of γ.*

(i) Uniqueness (tameness). *γ is* tame *when it admits a* unique *companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).*

(ii) Existence and closure. *A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and* if it exists, the lift is itself a loop *([21, Cor. 5.13]; pinned against the arXiv text; the earlier citation to Cor. 5.22 is superseded).*

(iii) Winding number. *For an untwisted tame loop with even circular signature σ^c, the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).*

7 Slice preservation and symmetries

With the ring in hand, we check that $\zeta_{\mathbb{O}^*}$ actually lives in it, and record the symmetry that will organize its zeros. Slice preservation says $\zeta_{\mathbb{O}^*}$ keeps each slice inside itself; together with slice regularity this makes it a section of $\mathcal{R}$. The symmetry is the action of the automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$, which fixes the real axis and rotates the imaginary directions S^6, and $\zeta_{\mathbb{O}^*}$ is equivariant under it. This forces every non-real zero to fill a whole 6-sphere, the geometry the next section reads off.

Theorem 7.1 (Slice preservation)**.** *$\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.*

Proof. Fix $v \in S^6$ and $s \in \mathbb{C}_v^*$. If $s \in \mathbb{C}_v \setminus \mathbb{R}$ then $s = \sigma \pm \gamma v$ and $\zeta_{\mathbb{O}^*}(s) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma \pm i\gamma)) \in \varphi_v(\mathbb{C}^*) = \mathbb{C}_v^*$ (Definition 4.1(i)). If $s \in \mathbb{R}\setminus\{1\}$ then $\zeta_{\mathbb{O}^*}(s) = \zeta(s) \in \mathbb{R} \subseteq \mathbb{C}_v^*$; while $\zeta_{\mathbb{O}^*}(1) = \infty \in \mathbb{C}_v^*$ and $\zeta_{\mathbb{O}^*}(\infty) = 1 \in \mathbb{C}_v^*$. In every case the value remains in $\mathbb{C}_v^*$. $\square$

Theorem 7.2 ($\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$)**.** *$\zeta_{\mathbb{O}^*} \in \mathcal{R}$.*

Proof. On each slice the restriction is the compactified classical zeta read through the isomorphism φ_v, namely $\zeta_{\mathbb{O}^*}|_{\mathbb{C}_v^*} = \varphi_v \circ \zeta_{\mathbb{C}^*} \circ \varphi_v^{-1}$, holomorphic into $\mathbb{C}_v^*$ away from the single pole. Equivalently $\zeta_{\mathbb{O}^*} = \mathrm{ext}(\zeta)$ is the slice-regular extension of the holomorphic, conjugation-symmetric ζ on $\mathbb{C} \setminus \{1\}$ (Theorem 3.4, [5, Thm. 5.1.5]); hence $\zeta_{\mathbb{O}^*}$ is slice regular on $\mathbb{O}^* \setminus \{1, \infty\}$ (slice regularity is slicewise holomorphy, Definition 3.1). By Theorem 7.1 it is slice preserving, $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$. A slice-regular, slice-preserving function $\mathbb{O}^* \to \mathbb{O}^*$ is by definition a section of $\mathcal{R} = \mathcal{O}_{\mathbb{O}^*}$ (Definition 5.1); the regular product on such sections restricts on each slice to the ordinary product (Theorem 5.2, Wang [16, Rem. 2.11]). Its zeros (Theorem 8.2) make $\zeta_{\mathbb{O}^*}$ a nonunit. $\square$

Definition 7.3 (The automorphism group G_2)**.** $G_2 = \mathrm{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$-algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\mathrm{Im}\,\mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g\cdot(\sigma+\gamma v) = \sigma+\gamma\, g(v)$.

Theorem 7.4 (G_2-equivariance)**.** *$\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.*

Proof. For $s \in \mathbb{R}$, $g \cdot s = s$ and $\zeta_{\mathbb{O}^*}(s) \in \mathbb{R}$ is fixed, so both sides equal $\zeta_{\mathbb{O}^*}(s)$. For a non-real $s = \sigma + \gamma v$ ($v \in S^6$, $\gamma > 0$), g carries the slice $\mathbb{C}_v$ to the slice $\mathbb{C}_{g(v)}$ by the isometric relabelling $\varphi_{g(v)} = g \circ \varphi_v$ (Definition 7.3). Since $\zeta_{\mathbb{O}^*}$ is defined slicewise by the same $\zeta_{\mathbb{C}^*}$ read through φ_v (Definition 4.1(i)),

$$\zeta_{\mathbb{O}^*}(g \cdot s) = \varphi_{g(v)}\big(\zeta_{\mathbb{C}^*}(\sigma + i\gamma)\big) = g\big(\varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))\big) = g \cdot \zeta_{\mathbb{O}^*}(s).$$

$\square$

Theorem 7.5 ([3])**.** *G_2 acts transitively on S^6 with stabilizer $\mathrm{SU}(3)$: $\mathrm{SU}(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.*

Remark 7.6 (The G_2-action extends to $\mathbb{O}^* = S^8$). The action above is stated on $\mathbb{O}$, but $\mathcal{H}_1$ (Part 3) runs on the compactified $\mathbb{O}^* = S^8$. Each $g \in G_2$ fixes the real axis pointwise (an automorphism fixes 1, hence all of $\mathbb{R}$) and acts on $\mathrm{Im}\,\mathbb{O} = \mathbb{R}^7$ as an element of $SO(7)$; in particular g is a linear isometry of $\mathbb{R}^8 = \mathbb{O}$, hence proper, and therefore extends uniquely to a homeomorphism of the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ fixing the point at infinity $\infty = N$ (the one-point compactification is functorial on proper maps; in Lean, `OnePoint`). Thus $G_2 \curvearrowright \mathbb{O}^*$ fixes the compactified real axis $\mathbb{R} \cup \{\infty\}$, a great circle through N, pointwise, and acts on each direction-sphere S^6 exactly as in Theorem 7.5. This is the action on which $\mathcal{H}_1$ is built, with $\mathcal{R}$ its ring of invariants over $\mathbb{O}^*$.

8 Zero geometry and the equivalence

This section is the bridge the whole paper is built to cross. It shows that the zeros of $\zeta_{\mathbb{O}^*}$ are exactly the classical nontrivial zeros, now spread out: each nontrivial zero ρ of ζ becomes a whole 6-sphere of zeros of $\zeta_{\mathbb{O}^*}$, its residue-$\mathbb{C}$ zero-sphere, sitting at real part $\operatorname{Re}\rho$. The Riemann Hypothesis, that every ρ has real part $\frac{1}{2}$, becomes the statement that all these 6-spheres share the same real part, that they are concentric. From here on we work with the spheres.

Theorem 8.1 (Zero Equivalence Theorem). *Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,*

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \quad \Longleftrightarrow \quad \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.7 these are exactly the nontrivial zeros of the classical ζ.

Proof. By Definition 4.1(i), $\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$. The slice identification $\varphi_v : \mathbb{C}^* \to \mathbb{C}_v^* = S_v^2$ is an isomorphism of slice Riemann spheres (the $\mathbb{R}$-algebra isomorphism $\mathbb{C} \to \mathbb{C}_v$ of Definition 2.4, extended by $\varphi_v(\infty) = N$; GPV [17]). An isomorphism is injective and sends 0 to 0, so $\varphi_v(z) = 0 \iff z = 0$; with $z = \zeta_{\mathbb{C}^*}(\rho)$ this gives both implications. (Slice preservation, Theorem 7.1, is what places the value in $\mathbb{C}_v^*$ so that φ_v^{-1} applies on the non-real side.) The final clause is Lemma 1.7. ☐

Theorem 8.2 (Nontrivial zeros are infinitely many disjoint 6-spheres). *For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set*

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma\, S^6 \subset \mathbb{O}.$$

Then:

(i) S_ρ is *a 6-sphere, the round sphere of radius γ centred at the real point σ, equivalently the G_2-orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser* SU(3)*, Theorem 7.5, Baez [3]).*

(ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar\rho}$;

(iii) the nontrivial-zero set is the disjoint union $Z^{\mathrm{nt}}_{\mathbb{O}^*} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar\rho\}$, with distinct pairs giving disjoint spheres;

(iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).

Proof. (i) The map $v \mapsto \sigma + \gamma v$ is the affine map $x \mapsto \sigma + \gamma x$ of $\operatorname{Im}\mathbb{O} \cong \mathbb{R}^7$ restricted to S^6; with $\gamma > 0$ its image is the round 6-sphere of radius γ centred at σ. By G_2-equivariance (Theorem 7.4) and transitivity of G_2 on S^6 (Theorem 7.5), $G_2 \cdot (\sigma + \gamma v_0) = \{\sigma + \gamma\, g(v_0) : g \in G_2\} = S_\rho$. So S_ρ *is* S^6, scaled by γ, translated to σ, realized as a single G_2-orbit; no embedding or diffeomorphism is invoked.

(ii) For any $v \in S^6$, $\zeta_{\mathbb{O}^*}(\sigma+\gamma v) = \varphi_v(\zeta_{\mathbb{C}^*}(\rho)) = \varphi_v(0) = 0$ by the Zero Equivalence Theorem 8.1. And $S_{\bar\rho} = \{\sigma + \gamma(-v) : v \in S^6\} = S_\rho$, since $v \mapsto -v$ is a bijection of S^6.

(iii) For "$\subseteq$": a non-real zero $s = \sigma + \gamma v$ of $\zeta_{\mathbb{O}^*}$ forces $\zeta_{\mathbb{C}^*}(\sigma + i\gamma) = 0$ by Theorem 8.1, so $s \in S_\rho$ with $\rho = \sigma + i\gamma$ nontrivial (Lemma 1.7); "$\supseteq$" is (ii). Disjointness: if $\sigma + \gamma v = \sigma' + \gamma' w$, uniqueness of the real/imaginary parts gives $\sigma = \sigma'$ and $\gamma v = \gamma' w$, whence $\gamma = \gamma'$ (taking norms, $|v| = |w| = 1$) and $\{\rho', \bar\rho'\} = \{\rho, \bar\rho\}$; so two spheres coincide or are disjoint.

(iv) There are infinitely many spheres because ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). ☐

Theorem 8.3 (The equivalence: concentricity $\Leftrightarrow$ RH)**.** *The following are equivalent:*

(a) the Riemann Hypothesis;

(b) the nontrivial-zero 6*-spheres* $\{S_\rho\}$ *of* $\zeta_{\mathbb{O}^*}$ *(Theorem 8.2) are* concentric*: they share a single common centre, necessarily a point of the real axis.*

When they hold, the common centre is $\frac{1}{2}$*.*

Proof. Each S_ρ is centred at the real point σ_ρ, where $\rho = \sigma_\rho + i\gamma_\rho$ (Theorem 8.2(i)).

$(a) \Rightarrow (b)$: under RH every $\sigma_\rho = \frac{1}{2}$, so all spheres share the centre $\frac{1}{2}$.

$(b) \Rightarrow (a)$: suppose the spheres share a common centre $c \in \mathbb{R}$. By the functional equation (Theorem 1.1), if ρ is a nontrivial zero then so is $1 - \bar{\rho} = (1 - \sigma_\rho) + i\gamma_\rho$, whose sphere $S_{1-\bar{\rho}}$ is centred at $1 - \sigma_\rho$. Concentricity forces $\sigma_\rho = c = 1 - \sigma_\rho$, hence $\sigma_\rho = \frac{1}{2}$; as ρ was arbitrary this is RH, and the common centre is $\frac{1}{2}$. □

Part III

The Categorical Construction and the Concentricity Theorem

Part 2 revealed that ζ carries a special geometry: spread onto the octonions, its zeros are concentric exactly when the Riemann Hypothesis holds. That raises the question this part answers. What general properties of a section of the ring $\mathcal{R}$ of slice-preserving functions force its residue-$\mathbb{C}$ zeros to be concentric? Following that question leads to four conditions, C1–C4, and to the whole class of functions satisfying them, which we call the A-sections. The main result, the Concentricity Theorem, is that the residue-$\mathbb{C}$ zeros of any A-section are concentric. The octonionic zeta is one A-section among many, and the Riemann Hypothesis is one of its corollaries.

From the section's own data we read one distinguished exponential Möbius transformation of the Riemann sphere, presented at its two fixed points by the Euler product and the Weierstrass product; orbit–stabilizer carries that single transformation across the projective base. The resulting map to SphereWorld is its geometric projection, while the same action on normalized A-value states assembles the genuine functor $\mathcal{A}_A : \mathcal{B} \to \mathbf{Grpd}$. Its Grothendieck construction $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ gathers every state and every value-transport into one total object. The zero-sphere system is the inverse-image subdiagram $\mathcal{R}_A$, included by $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$; the relative orbit–stabilizer comparison makes $\int_{\mathcal{B}} \mathcal{R}_A$ transitive and hence connected. The component colimit therefore collapses to a singleton, and its A-specific real-value readout is one centre $c \in \mathbb{R}$. The concentricity proof takes place in this residue-$\mathbb{C}$ inverse-image system inside the ambient total.

9 The geometric worlds and the categorical construction

This section assembles the parts the detector runs on. We begin with the compactified octonions $\mathbb{O}^* = S^8$ and their north pole N, the point at infinity where a section sends its pole. We then describe what a slice-preserving section does on each slice, one holomorphic stem repeated in every direction, which is why its non-real zeros fill whole 6-spheres. We set up the two geometric viewpoints that carry this data, the octonionic world $\mathcal{H}_1$ and the slice-value world $\mathcal{S}_2$. And we close with the projective base $\mathcal{B}$, the A-section functor over it, and its Grothendieck total $\mathcal{T}_A$, the object the next section reads.

The compactified octonions and the point at infinity

Definition 9.1 (The compactified octonions $\mathbb{O}^* = S^8$)**.** Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$. Concretely (GPV [17], Prop. 4.1 and Thm. 4.2, for $m = \dim_{\mathbb{R}} K \in \{4, 8\}$): the inverse stereographic projections from the north and south poles,

$$f(x+Iy) = \left(\tfrac{2(x+Iy)}{1+x^2+y^2},\ \tfrac{-1+x^2+y^2}{1+x^2+y^2}\right), \qquad h(x+Iy) = \left(\tfrac{2(x-Iy)}{1+x^2+y^2},\ \tfrac{1-x^2-y^2}{1+x^2+y^2}\right),$$

give a conformal atlas $\{(K, f), (K, h)\}$ endowing S^m with the structure of a slice (quaternionic/octonionic) Riemann manifold, with transition map $h^{-1} \circ f(q) = \bar{q}/q^2 = 1/q$, itself slice regular and slice preserving. The adjoined point is the north pole N; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`, $\mathbb{O}^* =$ `OnePoint` $\mathbb{O}$ with $\infty =$ `OnePoint.infty`, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$, the point at infinity $\infty = N$. The two views $\mathcal{H}_1$ and $\mathcal{S}_2$ are built on these compactified values (Definition 9.3); the distinct projective base $\mathcal{B}$ and direction-indexed sphere world used in the construction below are introduced in Definition 9.4.

There is one geometric north pole $N = \infty$, seen from two categorical points of view. In $\mathcal{H}_1$ this point of $\mathbb{O}^*$ is a G_2-fixed object: its own connected component, with $\mathrm{Aut}(N) = G_2$, carrying no universal property. In the slice-value world $\mathcal{S}_2$ the same geometric point is represented by the object

$$\mathfrak{n} \;:=\; \infty = N \text{ regarded as an object of } \mathcal{S}_2,$$

the common point at infinity of the slice Riemann spheres, fixed by band and direction alike. The two representatives belong to *different* categorical viewpoints and are related by the section through C1: the pole of A has value ∞, so the section carries the pole point to $\mathfrak{n}$. The residue field of a closed zero sorts the picture: residue-$\mathbb{R}$ (the G_2-fixed locus $\mathbb{R} \cup \{N\}$, classically trivial) versus residue-$\mathbb{C}$ (the 6-sphere orbits, classically nontrivial; the dictionary is Part 2 and Theorem 8.2). The later total construction uses this one shared compactified witness N, the same for every zero index.

What a slice-preserving section does

Definition 9.2 (The section's slice data and equivariance)**.** Let $A \in \mathcal{R}$. By the I-independent lift (Definition 6.4; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$-valued components,

$$A(x+Iy) = F_1(x+iy) + I\,F_2(x+iy), \qquad F_1, F_2\ \mathbb{R}\text{-valued},$$

the same for every $I \in S^6$. Three consequences used below:

(i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I: the value at a point lies on that point's own slice sphere (Definition 5.1).

(ii) *G_2-equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I)\,F_2 = g \cdot \big(F_1 + I\,F_2\big) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2-fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i). (For $\zeta_{\mathbb{O}^*}$ this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

(iii) *Blindness to the sphere direction.* On a G_2-orbit $S_{(\sigma,\gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

A slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2-morphisms already present in both worlds.

Condition C4, infinitely many residue-$\mathbb{C}$ zeros, matches the class to its infinite structure (the infinite Euler family of C2, the infinite Weierstrass divisor of C3) and gives the collapse its force: the degenerate fibre of the transport is an infinite family, so that whether it lies in one connected component of $\mathcal{T}_A$ (Definition 9.4) is an infinite alignment, decided by Theorem 10.7.

Two geometric viewpoints

Definition 9.3 (The octonionic and slice-value worlds)**.** Recall the *translation groupoid* of an action $G \curvearrowright X$: its objects are the points of X and its morphisms $x \to y$ are the elements $g \in G$ satisfying $g \cdot x = y$. Thus $\pi_0(G \ltimes X) = X/G$ (Quillen [22, §1]; in Lean, `CategoryTheory.ActionCategory`).

The octonionic world is

$$\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*.$$

Its components are the G_2-orbits: the compactified real fixed locus $\mathbb{R} \cup \{N\}$ and, for each $\sigma \in \mathbb{R}$ and $\gamma > 0$, the residue-$\mathbb{C}$ sphere $S_{(\sigma,\gamma)} = \{\sigma + \gamma v : v \in S^6\}$, labelled by its real centre σ and radius γ (Baez [3]; Theorem 7.5).

The slice-value world $\mathcal{S}_2$ organizes the same compactified values by their slice decomposition. Every slice contains the real axis, and

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \qquad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

Its objects are the points of $\mathbb{O}^*$. Its generating paths are the direction arrows induced by G_2 and the band phases $z \mapsto e^{I\theta} z$ within a compactified slice; the Lean object `S2` is the quotient path category by the direction-composition relations. In particular 0 and N are shared values, not new objects indexed by a zero.

These two geometric views hold the octonionic and slice-value geometry drawn on below.

The base of the degenerate set and the Grothendieck construction

Definition 9.4 (The projective base, genuine section functor, and total category)**.** The base is the projective action groupoid

$$\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \mathrm{OnePoint}(\mathbb{R}).$$

Its objects are points of the compactified real projective line and its arrows are projective transformations together with their transport equations.

The matrix construction makes explicit why the analytic object is simultaneously a function and a group element. Let

$$C = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}, \qquad M(z) = \begin{pmatrix} e^z & 0 \\ 0 & 1 \end{pmatrix}, \qquad D(z) = \left[C\, M(z)\, C^{-1}\right] \in \text{Möb}(\mathbb{C}^*).$$

The matrix C is the Cayley map $w \mapsto (w - i)/(w + i)$. Matrix multiplication and $e^{z+w} = e^z e^w$ give

$$M(z+w) = M(z)M(w), \qquad D(z+w) = D(z)D(w).$$

Thus exponentiation is the homomorphism which turns addition of logarithmic lifts into composition of the actual slice Möbius maps. In the Lean source these are the chain `diagonalGL`, `diagonalGLHom`, `expUnit`, and `diskExpAction`, with their multiplication laws. Conditions C1–C3 select the A-specific logarithm λ_A of the continued nonzero pole factor and hence the one element

$$D_A = D(\lambda_A) = \left[C \operatorname{diag}(e^{\lambda_A}, 1) C^{-1}\right].$$

The Euler and Weierstrass expressions are the two presentations of this same multiplier. The function/group-element passage is the commuting dictionary

$$\begin{tikzcd} z \arrow[r, "\exp"] \arrow[d, mapsto] & e^z \in \mathbb{C}^\times \arrow[d, mapsto] \\ M(z) = \operatorname{diag}(e^z, 1) \arrow[r, "C(-)C^{-1}"'] & D(z) \in \text{Möb}(\mathbb{C}^*). \end{tikzcd}$$

The common Euler–Weierstrass multiplier can be followed directly through this dictionary. Write p_A for the pole of A and $g_A =$ `distinguishedPoleFactor A`. On the punctured pole chart, conditions C2 and C3 give the two identities

$$g_A(z) = (z - p_A) \exp\left(\sum_{p \in A.\iota} \ell_p(z)\right),$$
$$g_A(z) = z^{m_A} R_{\text{fac},A}(z) e^{h_A(z)} \prod_{n \geq 0} \mathcal{E}\big(z; q_{A,n}\big).$$

Condition C1 continues this same factor through the simple pole and proves $u_A = g_A(p_A) \neq 0$. Hence $u_A \in \mathbb{C}^\times$, its logarithm $\lambda_A = \log u_A$ satisfies $e^{\lambda_A} = u_A$, and the distinguished element is

$$D_A = D(\lambda_A) = \left[C \begin{pmatrix} u_A & 0 \\ 0 & 1 \end{pmatrix} C^{-1}\right].$$

This calculation also gives its two fixed boundary points. The diagonal matrix $\operatorname{diag}(u_A, 1)$ fixes 0 and ∞ in the standard projective chart; conjugation by C carries those points to the projective zero and the shared north point N. Therefore

$$D_A(0) = 0, \qquad D_A(N) = N,$$

which are the declarations

`ASection.distinguishedDiskAction_fixes_cayley_zero`,

`ASection.distinguishedDiskAction_fixes_cayley_N`.

The Euler expression at 0 and the Weierstrass expression at N are consequently the two boundary readings of this one matrix-built function.

The GPV lift records how that function varies between its boundary presentations. Let $X, Y \in \mathcal{B}$. A projective-base transport from X to Y carries three continuous tapes: a compactified domain path δ^*, a nonzero value path v, its compactification v^*, and a logarithmic lift Γ. Their defining equations are

$$\begin{aligned} \delta^*(0) &= \mathrm{cayleyCoord}(X), & \delta^*(1) &= \mathrm{cayleyCoord}(Y), \\ v^*(t) &= A^*\big(\delta^*(t)\big), & e^{\Gamma(t)} &= v(t), \\ \Gamma(1) - \Gamma(0) &= 2\pi i k, & k &\in \mathbb{Z}. \end{aligned}$$

These are precisely the fields of `ASection.GpvTransport`. At each t the lift equation passes through the matrix dictionary:

$$\begin{tikzcd} \Gamma(t) \arrow[rr, "\exp"] \arrow[d, "D"'] & & v(t) \in \mathbb{C}^\times \arrow[d, "u \mapsto [C\,\mathrm{diag}(u,1)C^{-1}]"] \\ D(\Gamma(t)) \arrow[rr, equal] & & \big[C\,\mathrm{diag}(v(t),1)C^{-1}\big]. \end{tikzcd}$$

Thus every point of the lift is already an action of the same kind as D_A. Taking real parts and norms in the lift equation gives

$$\operatorname{Re}\Gamma(t) = \log\|v(t)\|.$$

The right-hand side shows that the real level is continuous and depends only on the value tape. Any second continuous lift with the same initial rung equals Γ, while every lift of the same value tape has this same real part. Taking real parts in the winding equation gives

$$\operatorname{Re}\Gamma(0) = \operatorname{Re}\Gamma(1),$$

and exponentiating the full equation gives

$$D(\Gamma(0)) = D(\Gamma(1)).$$

The winding integer therefore records the vertical displacement of the logarithmic rung, and the real level and the endpoint Möbius action remain fixed.

Condition C2 now supplies the canonical arithmetic instance of this construction. Along an Euler half-space loop δ,

$$\Gamma_A(t) = \sum_{p \in A.\iota} \ell_p(\delta(t)), \qquad e^{\Gamma_A(t)} = A(\delta(t)),$$

with the prime index retained inside the complete summation. Local normal convergence makes Γ_A continuous, the zero-free half-space makes its value tape lie in $\mathbb{C}^\times$, and the GPV uniqueness theorem fixes the whole lift once $\Gamma_A(0)$ is chosen. Since $\delta(0) = \delta(1)$, the same prime sum occurs at both endpoints, so its checked winding is $k = 0$. At every instant the square `canonicalAsectionPresentation_euler_toNorth` carries this complete prime tape to the common north frame. Condition C3 supplies the Weierstrass boundary presentation there, and condition C1 identifies both presentations with the continued factor g_A. The prime sum, its value under A, its logarithmic level and winding, and the matrix action it generates are therefore simultaneous coordinates of the distinguished element.

The projective action is read in the direction-indexed sphere-world groupoid. Its objects are directions $I \in S^6$ equipped with their compactified slice spheres, and its morphisms carry the direction and Möbius legs. A normalized value state pairs such a direction with a point of its slice. To position D_A over the whole base, fix the canonical orbit representative $o_b : N \to b$. Every arrow $f : x \to y$ has the unique residual north-stabilizer factor

$$r_f = o_y^{-1} f\, o_x, \qquad f = o_y r_f o_x^{-1}.$$

The object frame and the full arrow element are therefore

$$a_A(b) = C(o_b)D_A, \qquad a_A(f) = a_A(y)\, C(r_f)\, a_A(x)^{-1}. \tag{OS}$$

Here $C(-)$ denotes the multiplicative Cayley-projective homomorphism, and $f \mathbin{⨾} g$ denotes composition in diagrammatic order: f first, then g. The fixed orbit representatives force r_f uniquely; $r_{1_x} = 1$ and $r_{f⨾g} = r_g r_f$. These identities, together with multiplicativity of $C(-)$, prove the identity and composition laws directly from the fixed representatives.

Equation (OS) is the orbit–stabilizer well-definedness of every functor in this paper, and it is the only device used to obtain one. Each is built the same way: the distinguished element D_A, carried to every object by the Cayley map along the fixed orbit representatives. An object is positioned by $a_A(b) = C(o_b)D_A$; an arrow is positioned by $a_A(f)$, which is nothing but its residual stabilizer factor r_f conjugated between the frames at its two ends. Because r_f is forced and $C(-)$ is a homomorphism, the functor laws are not separate obligations — they are the two identities just displayed. We refer to (OS) by name throughout, and every later positioning statement is an instance of it rather than a new hypothesis.

The preceding construction gives nine checkable readings on the projective base:

- (B1) the C2 Euler and C3 Weierstrass formulas present the one C1-continued factor g_A;
- (B2) $u_A = g_A(p_A)$ and $\lambda_A = \log u_A$ generate the one matrix action D_A;
- (B3) the diagonal calculation fixes the two boundary points 0 and N;
- (B4) the Euler lift retains the complete prime-indexed sum $\sum_{p \in A.\iota} \ell_p$;
- (B5) exponentiation identifies the lift pointwise with the A-value and with its Möbius matrix;
- (B6) the initial rung determines the unique tame continuous lift;
- (B7) its real level is $\log\|A\|$, continuous and independent of the lift branch;
- (B8) its winding lies in $2\pi i\mathbb{Z}$, preserves the endpoint action and real level, and equals zero on the C2 Euler loop;
- (B9) the unique orbit and stabilizer factors carry every instant of this action through the base, with identity and composition inherited from their factorization laws.

Each line is a consequence of the construction above: (B1)–(B3) come from the common pole factor and its matrix, (B4)–(B8) from its GPV lift, and (B9) from the projective orbit–stabilizer sweep.

The direction-and-Möbius projection is

$$A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld},$$

written `AsectionSlice A` in the Lean source. It records the slice sphere and its Möbius transport. The full A-section functor below additionally retains the normalized point and the value produced on that sphere.

The slice-preserving realization on $\mathbb{O}^*$ is the G_2-equivariant endofunctor $A_{\mathbb{O}} : \mathcal{H}_1 \to \mathcal{H}_1$. The normalized state world has functorial input and output eyes, and the round trip is the checked triangle

$$\begin{tikzcd} \text{StateWorld}_A \arrow[r, "\text{In}_A"] \arrow[dr, "\text{Out}_A"'] & \mathcal{H}_1 \arrow[d, "A_{\mathbb{O}}"] \\ & \mathcal{H}_1, \end{tikzcd} \qquad \text{Out}_A = \text{In}_A \mathbin{⨾} A_{\mathbb{O}}.$$

This equality is `AsectionState_input_then_equivariant`: it holds on arrows as well as objects, so the normalized input coordinates genuinely drive the realized A-values.

Positioning this round trip over $b \in \mathcal{B}$ forms a constrained graph. An object in the fibre consists of an input state x, the forced positioned state $a_A(b)\cdot x$, and the forced value $\mathrm{Out}_A(a_A(b)\cdot x)$. Thus both the positioned point and its value are determined by the input and the action. For $f : b \to b'$, the two legs of the orbit–stabilizer square transport the input and positioned faces, and its already-proved commuting equation makes the target graph well-defined. Reading that same square through In_A and Out_A gives the checked input and output naturality squares. Identity and composition of these graph transports are inherited from the unique factorization of r_f. Sweeping the whole constrained graph through the base therefore gives the genuine A-section functor

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd},$$

written `AsectionActionDiagram A` in the Lean source. Explicitly, on objects and on arrows:

- **On objects.** A base object b is sent to the groupoid of normalized A-section states positioned by its object frame $a_A(b) = C(o_b)D_A$. An object of that groupoid is one constrained graph (A) — a slice direction $I \in S^6$ and a normalized input coordinate u, together with the positioned coordinate $a_A(b)\cdot(I,u)$ and its realized A-value. An arrow of that groupoid is a single element $\gamma \in G_2$, acting on the slice direction and carrying the whole graph with it: the fibre is the action groupoid of G_2 on the states, and nothing else.
- **On arrows.** A base arrow $f : x \to y$ is sent to transport by its arrow element $a_A(f) = a_A(y)\,C(r_f)\,a_A(x)^{-1}$ of (OS). On a state this moves the input coordinate by the input leg $C(r_f)$ and the positioned coordinate by the positioned leg, the two horizontal legs of the square whose verticals are the frames at the two ends; on an arrow it is the identity on γ, since G_2 commutes past the positioning. Functoriality is the two forced identities $r_{1_x} = 1$ and $r_{f ⨾ g} = r_g r_f$ carried by the homomorphism $C(-)$, and is therefore not a separate obligation.

The nine base readings (B1)–(B9) pass through this constrained graph by the certified projective/GPV action squares. Three further readings are thereby realized in the generated states. First, condition C3 supplies the populated residue-$\mathbb{C}$ sphere states, represented by `residueActionState` and certified by `residueActionState_mem`. Second, condition C4 makes their population infinite through `c4_infinite`. Third, every such state carries its intrinsic real coordinate through `transportLevel`. These are the three output readings. Together with (B1)–(B9) they form the $9 + 3$ partition: nine properties of the one distinguished action on the base, followed functorially by three readings of the A-generated residue states in the total.

The total value-transport category is

$$\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A.$$

An object is a base position together with a normalized A-section value state in its fibre. A morphism is a base channel together with the compatible A-section transport supplied by A. The populated residue-$\mathbb{C}$ zero-sphere states are the degenerate-fibre outputs of this completed construction, produced by C3–C4. The associated diagram $\pi_0 \circ \mathcal{A}_A : \mathcal{B} \to \mathbf{Set}$ is the components diagram required by the Grothendieck/colimit readout (Lemma 10.3).

10 The concentricity theorem

We now have the worlds and the base, so this section defines the class of functions precisely and proves the theorem. Two facts about slice-preserving functions come first: the Weierstrass factorization that gives condition C3 its meaning, and the fact that a residue-$\mathbb{C}$ zero is a whole 6-sphere, a single connected component of the octonionic world. Then the categorical engine, Riehl's computation of the components of a Grothendieck construction as a colimit, which sorts the assembled total into the classes linked by the section's real-value transports. With these in hand we define an A-section by its four conditions C1–C4, state and prove the Concentricity Theorem, and record the corollary that carries its conclusion back into the language of classical zeros.

Proposition 10.1 (Slice-regular Weierstrass factorization; the content of C3)**.** *Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as*

$$A = q^m \, R \, S \, h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order-m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A, S is a factor vanishing exactly at its spherical (residue-$\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds including the case where either family of zeros is infinite.

Proof. This is the factorization theorem of Altavilla–de Fabritiis [2, Prop. 3.1, Thm. 3.2], whose primary factors are the slice-regular ones of Gentili–Vignozzi [10]; convergence of the infinite products is part of the cited statement. The pair R, S is the residue split named in C3: R carries the residue-$\mathbb{R}$ (classically trivial) zeros and S the residue-$\mathbb{C}$ (classically nontrivial) ones, the latter infinite in number by C4. □

Lemma 10.2 (Residue-$\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$)**.** *For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of the translation groupoid $\mathcal{H}_1$.*

Proof. A is slice-preserving, so on a sphere $\sigma + \gamma S^6$ ($\gamma > 0$) it has the form $A(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\,\beta(\sigma, \gamma)$ with α, β real (the real stem, [16]); since v is an imaginary unit, $\alpha + v\beta = 0$ forces $\alpha = \beta = 0$, independently of v. Hence a non-real zero occurs on a *whole* sphere $\sigma + \gamma S^6$ or at none of its points (the spherical-zero structure, [1]); real zeros are isolated real points (residue field $\mathbb{R}$). Each such sphere $\sigma + \gamma S^6$ is the G_2-orbit of any of its points (Theorem 7.5: G_2 acts transitively on S^6 with stabiliser SU(3)), and the connected components of $\mathcal{H}_1$ are exactly the G_2-orbits: in a groupoid two objects share a component iff a morphism joins them, and a morphism $x \to y$ of $\mathcal{H}_1$ is a $g \in G_2$ with $g \cdot x = y$. Therefore the residue-$\mathbb{C}$ zeros, the non-real ones, are exactly the 6-sphere components of $A^{-1}(0)$. □

Lemma 10.3 (π_0 of a Grothendieck construction)**.** *For a functor $F : \mathcal{B} \to \mathrm{Grpd}$, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:*

$$\pi_0\Big(\int_{\mathcal{B}} F\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}} (\pi_0 \circ F).$$

Proof. By Riehl's proof of Lemma 8.3.4, for any $X : \mathcal{C} \to \mathbf{Set}$ each arrow of the category of elements imposes exactly the condition that the corresponding elements be identified in every cone, so that $\pi_0(\mathrm{el}\, X) \cong \mathrm{colim}_{\mathcal{C}}\, X$ [23, p. 102]. Taking $X = \pi_0 \circ F$, the colimit performs exactly the identifications generated by the maps of F. In the Lean development this equivalence is `pi0GrothendieckEquiv`, built from the canonical cocone `pi0Cocone` and `colimit.desc` over Mathlib's `colimit_sound` and `colimit_eq`. □

Definition 10.4 (A-sections)**.** An *A-section* is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) with the following four properties.

(C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple, lies at a real point, and its value there is the point at infinity $\infty = N$.

(C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, $A = \exp\big(\sum_p \ell_p\big)$ for an *infinite* family $\{\ell_p\}$ of slice-preserving slice-regular functions each zero-free on Ω_0, the family summable *locally normally* on Ω_0 (the sense in which the cited Euler products converge; Theorem 1.2); in particular A is zero-free on Ω_0.

(C3) *Infinite Weierstrass factorization.* Over the *full divisor* of A, including the single pole of (C1), at the real point p_0, multiplicity -1, carried by the explicit left factor, on $\mathbb{O}^* \setminus \{p_0\}$,

$$(q - p_0)\, A \;=\; q^m\, R\, e^g \prod_n \mathcal{E}(\,\cdot\,; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the *nonzero* residue-$\mathbb{R}$ (real) zeros of A, vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue-$\mathbb{C}$ zero-spheres of A, and $\mathcal{E}(\,\cdot\,; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite, the product converging *locally normally* on $\mathbb{O}^* \setminus \{p_0\}$ (the convergence of Proposition 10.1). (The left factor makes "full divisor" literal: the divisor of a meromorphic section includes its single pole; classically, Hadamard factors $(s-1)\zeta(s)$.)

(C4) *Infinitely many residue-$\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$; [2]); residue-$\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Definition 10.5 (The residue subdiagram and its inclusion)**.** The *residue locus* of an A-section is $\{z : A(z) = 0,\ \operatorname{Im} z > 0\}$ (`CResidueZeroLocus`). The *residue subdiagram* $\mathcal{R}_A : \mathcal{B} \to \mathbf{Grpd}$ (`AsectionCResidueDiagram`) is the preimage of the states this locus selects at the north frame — a preimage taken in the total action groupoid, not of a set — with the witnessing base arrow carried as data. Explicitly, on objects and on arrows:

- **On objects.** A base object X is sent to the full subgroupoid of $\mathcal{A}_A(X)$ on the witnessed states: those x for which there are a north residue state x_N and a base arrow $g : N \to X$ with $F_A(g)(x_N) = x$. The witness travels with the object.
- **On arrows.** A base arrow $f : X \to Y$ is sent to the restriction of the ambient transport $F_A(f)$. It lands, because a witnessed state stays witnessed with the *same* north state and the composite witness $g \fatsemi f$ — that is the whole content, and it is functoriality of F_A and nothing further.

Its inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$ (`AsectionCResidueInclusion`) sends a witnessed object to its ambient state and is the identity on arrows; it is fibrewise fully faithful onto that inverse image, and its naturality squares retain the ambient A-section transport itself.

Lemma 10.6 (Transitivity of the C-residue total)**.** *For an A-section A, the total inverse-image groupoid*

$$\int_{\mathcal{B}} \mathcal{R}_A$$

is transitive: any two of its objects are joined by a morphism.

Proof. Let

$$P, Q \in \int_{\mathcal{B}} \mathcal{R}_A$$

be arbitrary. By the inverse-image definition, the fibre data of P and Q supply north residue states x_N, y_N, together with base arrows

$$g : N \longrightarrow P_{\text{base}}, \qquad h : N \longrightarrow Q_{\text{base}},$$

whose A-section transports recover the two given fibre states. Thus the arbitrary problem reduces to constructing a north endomorphism $k : N \to N$ and a fibre morphism

$$\varphi : F_A(k)(x_N) \longrightarrow y_N.$$

Once these are obtained, functorial transport along $g^{-1} \fatsemi k \fatsemi h$ gives an ambient Grothendieck morphism $P \to Q$, and the full inclusion $\int \iota_A$ established above returns that same morphism to $\int_{\mathcal{B}} \mathcal{R}_A$. It therefore remains only to construct this north comparison.

Write $F_A = \mathcal{A}_A$ for the A-section action functor, $D = D_A$ for its one distinguished Euler–Weierstrass disk action, and $C =$ cayleyProjective. There is one projective base and one north object N. The inverse-image construction above has already fixed the unique tame GPV 0-to-N tape and its unique orbit–stabilizer factorization; each zero index selects a state along that fixed base transport.

Because the supplied states x_N, y_N lie in the north fibre of the C-residue inverse image selected by the C3 zero-sphere system, there are C-residue indices n_1, n_2 and directions $I_1, I_2 \in S^6$ for which their positioned coordinates are the actual C3 coordinates

$$z_1 = \text{sphereZero}(n_1), \qquad z_2 = \text{sphereZero}(n_2).$$

The north object frame D uniquely determines their inputs:

$$u_i = D^{-1}(z_i), \qquad z_i = D(u_i).$$

This is where C3 enters the argument. It supplies the Weierstrass face of the distinguished action and the precise inputs on which that action is evaluated. The relative orbit–stabilizer comparison below supplies transitivity.

And now we can conjugate all of these points of view. The A-section round trip

$$\mathrm{Out}_A = \mathrm{In}_A \mathbin{⨾} A_{\mathbb{O}}$$

starts with a normalized slice state (I, u), retains its input coordinate u, and realizes its A-value equivariantly as the direction $I \in S^6$ varies. Its projective eye is the A-slice functor $A^{\mathrm{slice}} : \mathcal{B} \to$ SphereWorld, which carries the direction, the compactified slice sphere, and the Möbius leg that positions that sphere over the base. At the north object the orbit representative is $o_N = 1$; hence the positioned frame is the distinguished action itself,

$$a_A(N) = D_A, \qquad D_A = \left[C \begin{pmatrix} e^{\lambda_A} & 0 \\ 0 & 1 \end{pmatrix} C^{-1} \right], \qquad \lambda_A = \log g_A(p_A).$$

The common pole datum is therefore read directly in the north action. There is one construction here and not two: *Euler to north*, for each square. Euler presents at 0, and the same presentation arriving at N *is* Weierstrass — C1 continues g_A *through* the simple pole and the Weierstrass divisor is exact at N, so at N the two expressions are the same function. That is what makes the datum nonzero and puts (OS) at our disposal: $u_A = g_A(p_A) \neq 0$ (Definition 9.4). The nonvanishing is thus not a hypothesis imposed on D_A; it is what the continuation delivers at the point where the two presentations agree. It is also exactly what makes the positioned action a group action: $\mathrm{diag}(u_A, 1)$ has determinant $u_A \neq 0$, so its class lies in the projective group, and D_A is its conjugate by the Cayley map, hence invertible. Each object frame $a_A(b) = C(o_b)D_A$ is then a group element as well, and orbit–stabilizer applies to the action they generate. There is moreover one N here and not three: the north object of the projective base $\mathcal{B}$, the point at infinity shared by the slice Riemann spheres, and the north pole of $\mathbb{O}^* = S^8$ are the same point (Definition 9.1), so the one fixed 0-to-N tape has one north frame, at which $o_N = 1$.

The full functor $F_A = \mathcal{A}_A$ is the A-section action functor. It joins these views in one constrained graph: at N it records

$$(I, u) \longmapsto \big((I, u),\ D_A \cdot (I, u),\ A_{\mathbb{O}}(D_A \cdot (I, u))\big). \tag{A}$$

What F_A does is sweep the slice. The direction I ranges over S^6 and G_2 carries it; the distinguished element D_A positions the coordinate u in the slice it selects; and $A_{\mathbb{O}}$ realizes the value there. One element, swept over all directions, is the whole content of the functor.

Its well-definedness is (OS) and nothing else. On objects, F_A inherits the frame $a_A(b) = C(o_b)D_A$; on arrows, it inherits $a_A(f)$, so the identity and composition laws it must satisfy are exactly the two forced identities $r_{1_x} = 1$ and $r_{f⨾g} = r_g r_f$ transported by the homomorphism $C(-)$. This is why every rectangle later in the proof commutes without a separate argument: each one is (OS) read at a particular pair of objects. The same holds for the A-slice functor and for $\mathcal{R}_A$, which is why the two levels can be compared at all.

The inverse-image groupoid $\mathcal{R}_A(N)$ supplies the inputs occurring in this graph. Its residue condition identifies the positioned entries of x_N, y_N with the authored C3 coordinates $z_i = \mathrm{sphereZero}(n_i)$, and the graph equation then pulls each one back uniquely through the north action:

$$u_i = D_A^{-1} \cdot z_i, \qquad D_A \cdot u_i = z_i. \tag{P}$$

Accordingly, the inputs used in the comparison are the canonical preimages of the selected residue states in the inverse-image groupoid.

Thus u is the normalized input coordinate, I is the slice direction, $z = D_A \cdot u$ is the positioned coordinate, and the third entry is its realized A-value. On arrows, (OS) positions an arrow

$f : x \to y$ with input leg $C(r_f)$ as $a_A(f) = a_A(y)\, C(r_f)\, a_A(x)^{-1}$, while G_2 carries the slice direction. At a loop on N the frames at both ends are $a_A(N) = D_A$, and this reads as conjugation, $a_A(f) = D_A\, C(r_f)\, D_A^{-1}$; that is the form the relative loop takes at the end of the comparison. The coordinate comparison, the sphere normalization, and the pole-generated action are consequently three readings of a single F_A-state.

In particular, x_N and y_N are two evaluations of this same graph in the normalized north frame. They are not two constructions: each is Euler to north, run at its own input. Let $\iota_1, \iota_2 \in \int \mathcal{R}_A$ be arbitrary; by the inverse-image definition they differ in nothing but their inputs, and those inputs lie in the C-residue zero locus. One square apiece, the same square.

The rectangles used below are therefore the functorial arrows $F_A(k_{EN,1})$ and $F_A(k_{EN,2})$, both running on the one fixed tape from the common projective zero frame to the one north frame. Write

$$k_{EN,1}, k_{EN,2} : 0 \longrightarrow N$$

for their base components and

$$r_1 = \mathrm{stabilizerPart}(k_{EN,1}), \qquad r_2 = \mathrm{stabilizerPart}(k_{EN,2}).$$

These residual factors are forced. Orbit–stabilizer gives $k_{EN,i} = o_N\, r_i\, o_0^{-1}$, and `stabilizerPart_unique` says that any north-stabilizer element giving this same factorization equals r_i.

We now read the two runs as commuting action squares. Their vertical arrows are the object frames: $a_A(0)$ at the source of the fixed tape, and $a_A(N) = D_A$ at its north frame. Their horizontal arrows are the two legs of (OS): on top the input legs

$$R_1 = C(r_1), \qquad R_2 = C(r_2);$$

along the bottom the arrow elements $a_A(k_{EN,1})$ and $a_A(k_{EN,2})$. Read at $x = 0$, $y = N$, (OS) says that each square commutes:

$$a_A(k_{EN,1})\, a_A(0) = D_A\, R_1, \qquad a_A(k_{EN,2})\, a_A(0) = D_A\, R_2. \tag{S}$$

What $F_A(k_{EN,1})$ and $F_A(k_{EN,2})$ contribute is that they carry both horizontal legs together with these equations.

The fixed tape supplies the input at which the two runs are read: they leave the same source object, so they share its normalized input coordinate, which we write u_*. The C3 boundary readings identify the corresponding positioned outputs with the authored coordinates,

$$a_A(k_{EN,1}) \cdot \big(a_A(0) \cdot u_*\big) = z_1, \qquad a_A(k_{EN,2}) \cdot \big(a_A(0) \cdot u_*\big) = z_2, \tag{B}$$

and (P) rewrites each authored coordinate as $z_i = D_A \cdot u_i$. The two runs may therefore be displayed as

$$\begin{tikzcd}
u_* \arrow[r, "R_1 = C(r_1)"] \arrow[d, "a_A(0)"'] & u_1 \arrow[d, "D_A"] \\
a_A(0) \cdot u_* \arrow[r, "a_A(k_{EN,1})"'] & z_1
\end{tikzcd}
\qquad
\begin{tikzcd}
u_* \arrow[r, "R_2 = C(r_2)"] \arrow[d, "a_A(0)"'] & u_2 \arrow[d, "D_A"] \\
a_A(0) \cdot u_* \arrow[r, "a_A(k_{EN,2})"'] & z_2
\end{tikzcd}$$

Evaluating (S) at u_*, then reading the left side by (B) and the right side by (P), gives

$$D_A \cdot \big(C(r_1) \cdot u_*\big) = a_A(k_{EN,1}) \cdot \big(a_A(0) \cdot u_*\big) = z_1 = D_A \cdot u_1,$$
$$D_A \cdot \big(C(r_2) \cdot u_*\big) = a_A(k_{EN,2}) \cdot \big(a_A(0) \cdot u_*\big) = z_2 = D_A \cdot u_2.$$

D_A was shown above to be invertible, so cancelling it on the left yields the two input equations

$$C(r_1)\cdot u_* = u_1, \qquad C(r_2)\cdot u_* = u_2. \tag{I}$$

The north comparison is now algebra. Put

$$r = r_2 r_1^{-1},$$

the residual factor of the second run relative to the first:

$$\begin{aligned} C(r)\cdot u_1 &= C(r_2 r_1^{-1})\, C(r_1)\cdot u_* \\ &= C(r_2)\cdot u_* = u_2, \end{aligned} \qquad \begin{aligned} D_A\, C(r)\, D_A^{-1}\cdot z_1 &= D_A\, C(r)\cdot u_1 \\ &= D_A\cdot u_2 = z_2. \end{aligned}$$

The corresponding loop in the projective base is

$$k = k_{EN,1}^{-1} \mathbin{⨾} k_{EN,2} : N \longrightarrow N.$$

Composition of stabilizer parts is multiplication in the same order as the action, while inversion sends a stabilizer part to its inverse. Consequently

$$\mathrm{stabilizerPart}(k) = \mathrm{stabilizerPart}(k_{EN,2})\, \mathrm{stabilizerPart}(k_{EN,1})^{-1} = r_2 r_1^{-1} = r. \tag{R}$$

— equivalently, from the unique factorizations $k_{EN,i} = o_N r_i o_0^{-1}$, the first run inverted and followed by the second cancels the common 0-frame o_0, and $o_N = 1$. So (I) and (R) say that the input leg of $F_A(k)$ carries u_1 to u_2, and conjugating by D_A says its positioned leg carries z_1 to z_2. That is the coordinate half.

The other half is the direction, and it is where the sweep does the work. Each north state carries a slice direction $I_i \in S^6$, and G_2 acts *transitively* on the unit imaginary octonions (Theorem 7.5), so there is $\gamma \in G_2$ with $\gamma I_1 = I_2$. This is the step the ambient total cannot take. F_A sweeps the slice — I ranges over S^6 with G_2 carrying it, D_A positions the coordinate in the slice that selects, $A_{\mathbb{O}}$ realizes the value — and transitivity holds on what is swept. The inverse image is taken over exactly that output, which is why $\int \mathcal{R}_A$ is transitive where $\int_{\mathcal{B}} F_A$ is not.

The transported coordinate equality and this direction action are the two components of an arrow

$$\varphi : F_A(k)(x_N) \longrightarrow y_N \tag{Φ}$$

in the north action fibre.

It remains to return from the north fibre to the arbitrary objects P, Q. Their inverse-image data supply arrows

$$g : N \longrightarrow P.\mathrm{base}, \qquad h : N \longrightarrow Q.\mathrm{base}$$

and identifications

$$F_A(g)(x_N) = P.\mathrm{fiber}, \qquad F_A(h)(y_N) = Q.\mathrm{fiber}. \tag{N}$$

The base component of the required total arrow is

$$(g^{-1} \mathbin{⨾} k) \mathbin{⨾} h : P.\mathrm{base} \longrightarrow Q.\mathrm{base}. \tag{G}$$

Recall that a morphism $P \to Q$ of $\int_{\mathcal{B}} F_A$ is a base arrow $f : P.\mathrm{base} \to Q.\mathrm{base}$ together with a fibre arrow $F_A(f)(P.\mathrm{fiber}) \to Q.\mathrm{fiber}$. Functoriality gives $F_A\big((g^{-1} \mathbin{⨾} k) \mathbin{⨾} h\big) = F_A(h) \circ F_A(k) \circ F_A(g^{-1})$, and $F_A(g^{-1})(P.\mathrm{fiber}) = x_N$ by (N). So the composite carries $P.\mathrm{fiber}$ back along g^{-1} to x_N; then (Φ) carries the result along k to y_N; finally $F_A(h)$ carries that arrow forward to $Q.\mathrm{fiber}$, with

(N) identifying its two ends. This fibre arrow over (G) is a morphism from P to Q in the ambient Grothendieck construction. Since ι_A is full, it has a preimage in the residue total $\int \mathcal{R}_A$; faithfulness makes that preimage unique. Hence P and Q are joined. Since they were arbitrary, $\int \mathcal{R}_A$ is transitive.

Throughout the calculation z_1 and z_2 are arbitrary C3 coordinates. Their common real value is extracted from the singleton colimit in the next step. The comparison above rests on the relative stabilizer cancellation through the single common source tape and north frame. □

Theorem 10.7 (Concentricity). *Let A be an A-section. The infinitely many C-residue zero S^6 spheres transported by the A-section functor $\mathcal{A}_A : \mathcal{B} \to$ **Grpd**, from the projective great-circle base $\mathcal{B}$ to the A-generated normalized value groupoids, form exactly one colimit/component class, and that singleton class carries one real value*

$$c \in \mathbb{R}.$$

Consequently every populated residue-$\mathbb{C}$ zero 6-sphere has centre c, so the infinitely many populated residue-$\mathbb{C}$ zero spheres are concentric.

Proof. We consider a slice-preserving section A of the ring $\mathcal{R}$ on the compactified octonions $\mathbb{O}^*$. Slice preservation hands us a coherently identified family of normalized Riemann spheres, naturally organized by a functor. The hypotheses push in the same direction. Specifying the analytic object, the theory suggests exponential equations; and $\mathbb{O}^*$ comes from the Cayley–Dickson construction with only one real axis. The projective great circle that axis compactifies to is framed by $PGL(2,\mathbb{R})$: a projective coordinate frame in which $\exp(I\theta)$ lives, and the analytic sections of $\mathcal{R}$ themselves transform as Möbius transformations — self-transformations of the slice-preserving section. Frames and transformations compose, invert, and act: they are groupoids. The shape of the ring therefore determines the base $\mathcal{B}$ of Definition 9.4 (`GreatCircle.Base`).

Conditions C1–C4 now determine the distinguished action of the A-section. The exponential equations lead to the commuting triangle $\pi \circ E = \exp$ of the logarithm manifold (Theorems 6.1 and 6.2), giving the unique tame lift of an exponential base (Theorem 6.5, Proposition 6.6). In the matrix dictionary of Definition 9.4, every instant of a lift Γ is the actual group element

$$D(\Gamma(t)) = \left[C \begin{pmatrix} e^{\Gamma(t)} & 0 \\ 0 & 1 \end{pmatrix} C^{-1} \right].$$

This element is also the slice function $w \mapsto D(\Gamma(t))(w)$. Hence the analytic and symbolic readings are two faces of the same object: the exponential equation proves which matrix acts, and matrix multiplication proves how those actions compose.

Definition 9.4 has already constructed the complete prime-indexed GPV tape beside the matrix element it generates, including its exponential square, uniqueness, continuous real level, winding equation, and transport to the north frame. Condition C3 supplies its Weierstrass presentation at N (Proposition 10.1). C1 continues both presentations to the same nonzero pole factor, whose logarithm selects D_A. Consequently the Euler and Weierstrass faces are the two boundary presentations of the same matrix-built function/group element at its fixed points 0 and N. The degenerate fibre of Lemma 6.3, with its real level and winding heights, is this action's own bookkeeping.

Orbit–stabilizer now sweeps the element across the base. The canonical o_b positions D_A at each footpoint; the uniquely forced $r_f = o_y^{-1} f o_x$ supplies the residual north action; and the identity and multiplication laws for r_f give the functor laws. The G_2-equivariant realization (Theorem 7.4) applies the same positioned Möbius function to the normalized value states along the continuum of Riemann spheres. The result is the production A-section functor $\mathcal{A}_A =$ `AsectionActionDiagram A` of Definition 9.4; its states carry their authored inputs, positioned values, and realized A-outputs, and its arrows are the action's actual two-legged transports.

From the functor we form its total Grothendieck construction, $\mathcal{T}_A = \int_{\mathcal{B}} \mathcal{A}_A$ (Definition 9.4): an object is a base position together with a normalized value state in its fibre, a morphism is a base channel together with the compatible transport, and the whole is an action groupoid formed from the distinguished element. The residue-$\mathbb{C}$ zeros occur as states of this total categorical object. The distinguished element fixes 0 and N, and between them the upper half-space; at the north frame, where the positioning factor is trivial and the frame is the element itself, the equation $A = 0$ on that half-space selects the residue states, with Euler presenting at one end, Weierstrass at the other. The preimage of this selection under the action — the preimage of the total action groupoid, membership travelling by composition in the base — is the residue subdiagram $\mathcal{R}_A$ of Definition 10.5, included in $\mathcal{A}_A$ by ι_A. The restriction of the total object to this preimage sits over the same base.

This restriction carries the same unique fixed 0-to-N tape for every residue zero. At the north object the canonical positioning factor is the identity, so the frame is D_A itself. The checked north-frame and endpoint-fixation declarations identify its two fixed boundary points. The Euler and Weierstrass expressions are the two endpoint presentations of this one action, and the checked GPV continuation between them is unique once its initial rung is fixed. Moreover every projective arrow has the factorization

$$f = o_y\, r_f\, o_x^{-1}, \qquad r_f = \text{stabilizerPart}(f),$$

and `stabilizerPart_unique` identifies r_f as the unique north-stabilizer element in this factorization. Thus the inverse-image witness carries the already determined 0-to-N tape and its unique residual stabilizer part; a C3 zero index selects a state on that fixed base channel.

The fibrewise inverse image. For each base object X, an object of $\mathcal{R}_A(X)$ consists of an ambient state $x \in \mathcal{A}_A(X)$ together with a north residue state x_N and an arrow $g : N \to X$ satisfying

$$F_A(g)(x_N) = x. \tag{W}$$

This is the object property `IsCResidueState A X`. The fibre $\mathcal{R}_A(X)$ is the full subgroupoid of $\mathcal{A}_A(X)$ on precisely these objects. Its inclusion

$$\iota_{A,X} : \mathcal{R}_A(X) \hookrightarrow \mathcal{A}_A(X)$$

sends a witnessed object to its ambient state. For two such objects x, y, the induced map

$$\mathrm{Hom}_{\mathcal{R}_A(X)}(x, y) \longrightarrow \mathrm{Hom}_{\mathcal{A}_A(X)}(\iota_{A,X}x, \iota_{A,X}y) \tag{H}$$

is a bijection: every ambient arrow between the selected states is already an arrow of the full subgroupoid, and its ambient arrow determines it uniquely. Surjectivity of (H) is fullness of $\iota_{A,X}$; injectivity is faithfulness.

Transport of the inverse image. Let $f : X \to Y$. If x has witness (W), then $F_A(f)(x)$ has the same north state x_N and the composite witness $g \mathbin{⨟} f : N \to Y$, because

$$\begin{aligned} F_A(g \mathbin{⨟} f)(x_N) &= F_A(f)\big(F_A(g)(x_N)\big) \\ &= F_A(f)(x). \end{aligned} \tag{C}$$

This is the calculation recorded by `cResidue_lands`. It defines the restricted transport

$$\mathcal{R}_A(f) : \mathcal{R}_A(X) \longrightarrow \mathcal{R}_A(Y) :$$

on objects it uses the composite witness in (C), and on arrows it applies the ambient transport $F_A(f)$. Composition in the projective base therefore supplies closure of the inverse-image system under every A-section transport.

The natural inclusion. The fibrewise inclusions assemble over $\mathcal{B}$ in the square

$$\begin{tikzcd} \mathcal{R}_A(X) \arrow[r, "\mathcal{R}_A(f)"] \arrow[d, hook, "\iota_{A,X}"'] & \mathcal{R}_A(Y) \arrow[d, hook, "\iota_{A,Y}"] \\ \mathcal{A}_A(X) \arrow[r, "F_A(f)"'] & \mathcal{A}_A(Y). \end{tikzcd} \tag{Nat}$$

For an object x, both routes in (Nat) have underlying state $F_A(f)(x)$; for an arrow α, both routes have underlying arrow $F_A(f)(\alpha)$. The upper route additionally carries the residue witness $g \mathbin{⨾} f$ established in (C). Hence

$$\mathcal{R}_A(f) \mathbin{⨾} \iota_{A,Y} = \iota_{A,X} \mathbin{⨾} F_A(f). \tag{Inc}$$

and these equations for every f are the naturality of $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$.

The preimage now assembles over the base into a total of its own. Its objects are the positioned squares of the C-residue image and its morphisms are the value transports of $\mathcal{T}_A$ between them. Its inclusion is the total reading of ι_A,

$$\textstyle\int_{\mathcal{B}} \mathcal{R}_A \xhookrightarrow{\int \iota_A} \int_{\mathcal{B}} \mathcal{A}_A = \mathcal{T}_A$$

(`AsectionCResidueInclusionTotal` in the Lean source, the Grothendieck functoriality of ι_A applied to it). On objects it sends (X, x) to $(X, \iota_{A,X} x)$ and leaves the base position unchanged; on a Grothendieck morphism it leaves the base arrow unchanged and applies the fibre inclusion only to the compatible transport. Consequently equality after inclusion forces equality of the base legs and, by fibrewise faithfulness, equality of the transport legs: the total inclusion is faithful. Conversely, an ambient total morphism between two objects in the image already has a base leg f and a fibre morphism between selected target states; fibrewise fullness gives its preimage in $\mathcal{R}_A$, with the same f. Hence the total inclusion is full (`AsectionCResidueInclusionTotal_full` and `AsectionCResidueInclusionTotal_faithful`). It is therefore an isomorphism onto its image. The C-residue total is that image itself: it consists exactly of the selected objects together with the fixed tape and all ambient total morphisms between them.

Lemma 10.6 proves, in this exact inverse-image groupoid, that any two objects are joined by one Grothendieck morphism: the relative orbit–stabilizer face is its base component and the G_2 transport is its fibre component. Thus $\int_{\mathcal{B}} \mathcal{R}_A$ is connected. This is a groupoid-register statement: at this stage the established conclusion is connectedness; equality of real coordinates and the common centre arise from the colimit readout below.

By Remark 8.3.5 of [23], a category is nonempty and connected if and only if its π_0 is the singleton set. Condition C4 supplies nonemptiness — infinitely many residue-$\mathbb{C}$ zero-sphere states populate the preimage — and connectedness was just established. Therefore, by Lemma 10.3 (`pi0GrothendieckEquiv`),

$$\pi_0\Big(\int_{\mathcal{B}} \mathcal{R}_A\Big) \;\cong\; \operatorname*{colim}_{\mathcal{B}} (\pi_0 \circ \mathcal{R}_A)$$

collapses to a singleton κ. Here val is exactly the A-specific instantiation of that singleton equality at certified residue representatives. Their real coordinate is definitionally the project's level read

$$\mathrm{transportLevel}_A(n) = \mathrm{Re}\big(A.\,\mathrm{sphereZero}(n)\big).$$

Instantiating the singleton equality at the n-th and 0-th certified representatives is itself the val step:

$$\text{transportLevel}_A(n) = \text{transportLevel}_A(0).$$

The proof then sets $c = \text{transportLevel}_A(0)$. Every populated residue-$\mathbb{C}$ zero 6-sphere has real centre c, so the infinitely many residue-$\mathbb{C}$ zero spheres of the A-section are concentric. ☐

Corollary 10.8 (Translation to the classical framework)**.** *Under the residue-$\mathbb{C}$/classically-nontrivial-zero identification, the already-proved populated residue-$\mathbb{C}$ zero spheres are the classically nontrivial zero spheres and retain the common centre c supplied by Theorem 10.7.*

Proof. This is the translation of the C-residue population through Theorems 8.1 and 8.2. The centre comes directly from Theorem 10.7, while C4 and Lemma 10.2 supply infinitude and disjointness. ☐

11 Corollaries

Two corollaries close the argument. The first checks that the octonionic zeta is an A-section, verifying its four conditions C1–C4 one by one against the classical facts of Part 1; the Concentricity Theorem then applies to it directly, and its residue-$\mathbb{C}$ zero-spheres share one real centre c. The second corollary is the Riemann Hypothesis. The functional equation of ξ now pins that centre: it sends a zero of real part c to one of real part $1-c$, and applied to the common centre it forces $c = 1-c$, so $c = \frac{1}{2}$. Every nontrivial zero lies on the critical line.

Corollary 11.1 ($\zeta_{\mathbb{O}^*}$ is an A-section)**.** *$\zeta_{\mathbb{O}^*}$ is an A-section* (Definition 10.4)*: a section of $\mathcal{R}$ satisfying* (C1)–(C4).

Proof. $\zeta_{\mathbb{O}^*} \in \mathcal{R}$ by slice-preservation (Theorem 7.2). The verification of (C1)–(C4) cites the classical facts of the Riemann zeta function (Part 1) and the slice-regular zero-geometry that translates them:

- **(C1)** *Meromorphic continuation; simple pole.* Classical fact: ζ is meromorphically continued with a single simple pole at $s = 1$ (Theorem 1.1); the value of $\zeta_{\mathbb{O}^*}$ there is the point at infinity, $\zeta_{\mathbb{O}^*}(1) = \infty = N$ (Definition 4.1).
- **(C2)** *Infinite Euler product.* Classical fact: $\zeta = \prod_p (1-p^{-s})^{-1} = \exp(\sum_p \ell_p)$ on $\Re s > 1$, the family indexed by the infinitely many primes (Theorem 1.2).
- **(C3)** *Infinite Weierstrass factorization.* Supplied by the *slice-regular* Weierstrass factorization (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]): $(q-1)\,\zeta_{\mathbb{O}^*} = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n)$ (pole factor at $p_0 = 1$; classically, Hadamard factors $(s-1)\zeta(s)$), where $m = 0$ ($\zeta(0) = -\frac{1}{2} \neq 0$), R is the product over the nonzero residue-$\mathbb{R}$ zeros, the classically trivial zeros $-2, -4, \ldots$, honest real zeros of $\zeta_{\mathbb{O}^*}$, and $\{q_n\}$ enumerates its residue-$\mathbb{C}$ zero-spheres.
- **(C4)** *Infinitely many residue-$\mathbb{C}$ zeros.* Classical fact: ζ has infinitely many nontrivial zeros (Corollary 1.4; also Hardy, Theorem 1.5). By the residue dictionary below these are exactly the residue-$\mathbb{C}$ zeros, so $\{q_n\}$ is infinite.

After the (C1)–(C4) verification, the downstream centre-pinning uses the *functional equation* $\xi(s) = \xi(1-s)$, $\xi(\bar{s}) = \overline{\xi(s)}$ (Theorem 1.1), used in Corollary 11.2. The *residue dictionary*, a closed zero of $\zeta_{\mathbb{O}^*}$ is a real point (residue field $\mathbb{R}$, the classically *trivial* zeros) or a 6-sphere $S_{(\sigma,\gamma)}$ (residue field $\mathbb{C}$, the classically *nontrivial* zeros), is the slice-regular zero-geometry of Theorem 8.2 and [1]; it is what makes "residue-$\mathbb{C}$" and "nontrivial" the same notion. Thus (C1)–(C4) hold. ☐

Corollary 11.2 (Riemann Hypothesis)**.** *Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.*

Proof. By Corollary 11.1, $\zeta_{\mathbb{O}^*}$ satisfies (C1)–(C4). By Theorem 10.7, its populated residue-$\mathbb{C}$ value-transport colimit is the real singleton $\{c\}$ and its residue-$\mathbb{C}$ zero spheres are already concentric about c. By Corollary 10.8, these are the classically nontrivial zeros, so every nontrivial zero has real part c. The functional equation $\xi(s) = \xi(1-s)$ (Theorem 1.1) sends a zero of real part c to one of real part $1-c$; applied to the *common* centre it forces $c = 1-c$, whence $c = \frac{1}{2}$ (this is Theorem 8.3 read on $\zeta_{\mathbb{O}^*}$). Therefore every nontrivial zero has real part $\frac{1}{2}$, and by (C4) infinitely many lie on $\Re s = \frac{1}{2}$. $\square$

A word on why the argument takes the shape it does. No one can hold an 8-dimensional graph in the mind's eye, or draw the continuum of Riemann spheres the section lives on. The epigraph names the way through: no single viewpoint sees the whole, but when the viewpoints are multiplied and conjugated, a vision appears that none of them held alone. Each slice direction is one such viewpoint; the value-bearing A-section functor gathers all of them over the base, the total object holds them together, and the colimit conjugates them into a single answer. The concentricity we could not picture becomes one component we can compute.

Remark 11.3 (The image span of a fully faithful inclusion)*.* For any fully faithful functor $F : \mathcal{C} \to \mathcal{D}$ there is a canonical roof through the image,

$$\begin{tikzcd} & \mathrm{Im}(F) \arrow[dl, "F^{-1}"'] \arrow[dr, hook, "j"] & \\ \mathcal{C} & & \mathcal{D} \end{tikzcd}$$

the backward leg being an isomorphism of categories onto the image and the forward leg the canonical inclusion. It is a general fact, available here for ι_A and for $\int \iota_A$, and it is the shape a length-two zigzag argument would use. Theorem 10.7 obtains connectedness directly from transitivity at the total, by a single element. The image span records the corresponding length-two alternative.

Remark 11.4 (Two alternative routes)*.* Two other proofs of Theorem 10.7 are available. The first runs over the base. The restriction sits over $\mathcal{B}$ through its two projections,

$$\begin{tikzcd} \int_{\mathcal{B}} \mathcal{R}_A \arrow[rr, hook, "\int \iota_A"] \arrow[dr, "\pi_{\mathcal{R}}"'] & & \int_{\mathcal{B}} \mathcal{A}_A \arrow[dl, "\pi_{\mathcal{A}}"] \\ & \mathcal{B} & \end{tikzcd}$$

and one may connect residue states footpoint by footpoint: zigzags in the connected base, lifted through the fibres. That route travels through $\mathcal{B}$ and reassembles the value data leg by leg. The second is Thomason's homotopy colimit theorem [25]: the classifying space of a Grothendieck construction is homotopy equivalent to the homotopy colimit of the classifying spaces of its fibres, and reading connected components through that equivalence yields the same collapse of $\pi_0\big(\int_{\mathcal{B}} \mathcal{R}_A\big)$. The proof given here stays at the level of categories, where Lemma 10.3 suffices: the base route assembles what the action already supplies whole, the Thomason route is the topological shadow of the same computation, and both remain expository routes outside the Lean development.

References

[1] A. Altavilla and C. De Fabritiis, *$*$-logarithm for slice regular functions*, Rend. Lincei Mat. Appl. (arXiv:2106.04227); the slice-regular exponential $\exp_*$ (Def. 2.14, and Rem. 2.23 for the slice-preserving formula $\exp_*(f) = e^{f_0}(\cos f_1 + I \sin f_1)$) and the $*$-logarithm on never-vanishing functions (Def. 2.24, Thm. 6.17).

[2] A. Altavilla and C. De Fabritiis, *s-regular functions which preserve a complex slice*, Ann. Mat. Pura Appl. (arXiv:1801.01318); Prop. 3.1 and Thm. 3.2 factor a slice-regular function into a factor carrying exactly its (real and spherical) zeros and a never-vanishing factor, including the case of infinitely many zeros, the citable form of C3.

[3] J. C. Baez, *The octonions*, Bull. Amer. Math. Soc. **39** (2002), 145–205.

[4] C. Bisi and J. Winkelmann, *Invariants and Automorphisms for slice regular functions: the octonionic case*, arXiv:2411.16762 (2024); companion arXiv:2411.15896 in J. Noncommutative Geometry (EMS). Supplies the I-independent lift $\phi_I \circ F = A \circ \phi_I$ (the slice projection) and, in §3.2/§3.7/§14, the axially-symmetric-domain, slice-preserving, and central-divisor notions.

[5] F. Colombo, I. Sabadini, and D. C. Struppa, *Noncommutative Functional Calculus: Theory and Applications of Slice Hyperholomorphic Functions*, Progress in Mathematics **289**, Birkhäuser, 2011.

[6] R. Ghiloni and A. Perotti, *Slice regular functions on real alternative algebras*, Adv. Math. **226** (2011), 1662–1691.

[7] R. Ghiloni, A. Perotti, and C. Stoppato, *Singularities of slice regular functions over real alternative $*$-algebras*, Adv. Math. **305** (2017), 1085–1130 (slice-function and stem-function formalism, §1–§2; the slice-preserving $\Leftrightarrow$ intrinsic $\Leftrightarrow$ $\mathbb{R}$-valued-stem characterization used in Definition 6.4; §11: slice semiregular functions, Def. 11.1, Thm. 11.3).

[8] G. Gentili and D. C. Struppa, *A new theory of regular functions of a quaternionic variable*, Adv. Math. **216** (2007), 279–301.

[9] G. Gentili, C. Stoppato, and D. C. Struppa, *Regular Functions of a Quaternionic Variable*, Springer Monographs in Mathematics, Springer, 2013.

[10] G. Gentili and I. Vignozzi, *The Weierstrass factorization theorem for slice regular functions over the quaternions*, Ann. Global Anal. Geom. **40** (2011), 435–466.

[11] G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 1012–1014.

[12] J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000.

[13] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin. Akad. (1859).

[14] R. D. Schafer, *An Introduction to Nonassociative Algebras*, Academic Press, 1966.

[15] E. C. Titchmarsh (rev. D. R. Heath-Brown), *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.

[16] X. Wang, *On geometric aspects of quaternionic and octonionic slice regular functions*, arXiv:1512.01414v3.

[17] G. Gentili, J. Prezelj, and F. Vlacci, *Slice conformality and Riemann manifolds on quaternions and octonions*, Math. Z. **302** (2022), 971–994 (arXiv:2107.07892v2).

[18] The Stacks Project Authors, *The Stacks Project*, `https://stacks.math.columbia.edu`, 2024.

[19] The mathlib Community, *The Lean mathematical library* (`CategoryTheory.ActionCategory`, `CategoryTheory.ConnectedComponents`, `CategoryTheory.Comma`), `https://leanprover-community.github.io/mathlib4_docs/`.

[20] P. G. Goerss and J. F. Jardine, *Simplicial Homotopy Theory*, Modern Birkhäuser Classics, Birkhäuser, 2009 (Ch. I: simplicial sets and the nerve; Ch. IV: the bisimplicial engine for homotopy colimits).

[21] G. Gentili, J. Prezelj, and F. Vlacci, *On a continuation of quaternionic and octonionic logarithm along curves and the winding number*, J. Math. Anal. Appl. **536** (2024), no. 1, Paper No. 128219, DOI 10.1016/j.jmaa.2024.128219 (arXiv:2307.14047) (Rem. 2.1: the direction $I(q)$ has no continuous extension to $\mathbb{R}$; Def. 4.7: tame path = unique companion (Def. 4.20 for maps; Def. 5.2: tame/semi-tame at an obstruction parameter); Def. 5.11: the loop lift $\mathrm{pr}_1 \circ \Gamma = \gamma \circ \exp$; Cor. 5.13: a lift of a loop exists iff the signature lies in $\{0, -1\}$ on each obstruction interval, and is then a loop; Cor. 5.21: winding $= |\sigma^c|/2$).

[22] D. Quillen, *Higher algebraic K-theory: I*, in *Algebraic K-theory, I*, Lecture Notes in Math. **341**, Springer, 1973, 85–147 (Theorem A: a functor whose comma categories are contractible is final and induces an equivalence of classifying spaces).

[23] E. Riehl, *Categorical Homotopy Theory*, New Mathematical Monographs **24**, Cambridge University Press, 2014 (the nerve, classifying spaces, the two-sided bar construction, and homotopy colimits; Ch. 1, 4–6).

[24] E. Riehl, *Category Theory in Context*, Aurora: Dover Modern Math Originals, Dover Publications, 2016 (action groupoids and their orbits; Ex. 1.5.19).

[25] R. W. Thomason, *Homotopy colimits in the category of small categories*, Math. Proc. Cambridge Philos. Soc. **85** (1979), 91–109 (Thm. 1.2: the classifying space of the Grothendieck construction as a homotopy colimit).

[26] A. Vistoli, *Notes on Grothendieck topologies, fibered categories and descent theory*, in *Fundamental Algebraic Geometry*, Math. Surveys Monogr. **123**, AMS, 2005 (arXiv:math/0412512); §3.1 for the Grothendieck construction and categories fibered in groupoids.